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ConvNeXt-Mask2Former: A Semantic Segmentation Model for Land Classification in Remote Sensing Images

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***Abstract*—With the rapid development of remote sensing technology and deep learning, significant progress has been made in land classification of remote sensing images using semantic segmentation models. However, it still faces some deficiencies and challenges in terms of multi-scale features of images, the computational burden of high resolution images, and long-range dependencies. Therefore, in this paper, I design a semantic segmentation network based on encoder and decoder (Encoder-Decoder) architecture, using ConvNeXt as the encoder, i.e., the backbone network, and Mask2Former as the decoder, with the addition of an auxiliary segmentation module (ASM), and at the same time, an image enhancement pipeline is used to assist the training in the image preprocessing stage. Mask2Former is an architecture with good flexibility and generalization ability, which can be easily used in different visual tasks. At the same time, as a convolutional network without the addition of an attention mechanism, ConvNeXt has excellent feature extraction capabilities. I evaluate my model on the widely used high resolution image datasets Potsdam and Vaihinge and compare the results with the current mainstream semantic segmentation models, and the experimental results show that my model can be optimized for the accurate segmentation of land categories. Specifically, my model has an optimal mPA of 93.79% and mIoU of 86.80% on the Potsdam dataset, and an optimal mPA of 82.32% and mIoU of 90.05% on the Vaihinge dataset, which is both significantly higher than the existing models, and provide important references for land classification tasks and land use.**

Keywords-ConvNeXt; Mask2Former; semantic segmentation; remote sensing images; land classification

I. INTRODUCTION

In recent years, with the rapid development of remote sensing technology and computer vision field, semantic segmentation has played a key role in many fields, such as land classification and utilization, medical image analysis, automatic driving [1]and assessment, and emergency response of natural disasters. Semantic segmentation has important application value in remote sensing images. It is a fundamental task aimed at classifying each pixel in a remote sensing image to enable accurate recognition and analysis of categories in different tasks [2]. For example, the application of semantic segmentation in the field of land use is exactly one of the key tasks in remote sensing image processing. By semantic segmentation of remote sensing images, each pixel in the image can be assigned to different land use categories. Such as land classification, land change detection, land cover assessment.

Urban land classification is the systematic categorization and classification of urban land, aimed at providing a comprehensive understanding and effective management of the various types and uses of urban land. Classifying and categorizing urban land, for example, provides an understanding of the uses, functions, and characteristics of different areas. This helps urban planners, policymakers, and decision makers to better understand the distribution, current status of land use, and potential of land resources, so that they can formulate appropriate development strategies and planning programs to improve the efficiency of land use and promote sustainable urban development. There are many widely used urban land classification methods. In the early stage, experts mainly rely on traditional machine learning methods for land classification, such as Support Vector Machine (SVM) [3], Random Forest, and Decision Tree. These methods classify land by manually extracting features and constructing statistical models. The advantage is that they are simple and easy to implement and perform well for small-scale datasets. However, these methods usually rely on domain experts for the design of features and may have limitations in accuracy and generalization ability in complex land classification tasks.

With the rise of deep learning and the abundance of remote sensing image data, land classification methods at this stage mainly use deep learning algorithms, such as full convolutional neural FCN [4], DeepLabV3 [5], and Segformer [6]. These methods can automatically learn image features and achieve pixel-level classification without manual feature engineering. By using remotely sensed image data, including multispectral images, high resolution images, and radar images [7], deep learning methods are able to achieve more accurate land classification and segmentation results. FCN is one of the early deep learning methods, which is specifically used for image segmentation. However, FCN may not be able to fully recover all the details of the input image during the up-sampling process, especially when predicting edges and small objects, such as roads, buildings, and other small features, which are particularly important for land classification in high resolution remote sensing images and may be lost in the rough prediction. Moreover, FCN lacks a full understanding of the contextual relationships of features in remotely sensed images, but complex spatial relationships may exist between neighboring land classes.

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To overcome these shortcomings, subsequent research efforts have developed a series of improved architectures, such as the DeepLab family that introduces null convolution and SegFormer that integrates Transformer [8], which is designed to improve feature extraction, capture of contextual information, and fusion of multi-scale information. The DeepLabV3 network improves the model's ability to capture contextual information by null convolution for the capturing contextual information, but it may still fall short in accuracy in dealing with complex boundaries and small targets and uses a large number of null convolutions, which may lead to higher computational and storage requirements, especially in the case of high resolution of remote sensing images. SegFormer combines the advantages of Transformer with the efficiency of convolutional networks, but it may require a large amount of data and labelling to train, which may be a challenge for some remote sensing image datasets that may have issues such as high labelling costs and imbalance of labels, which may pose a challenge for applications of remote sensing image analysis where understanding the predictions of the model is important for such tasks.

Therefore, this paper addresses the above problem by designing a semantic segmentation network based on Encoder-Decoder architecture with ConvNeXt [9] as the encoder, i.e., the backbone network, Mask2Former [10] as the decoder, and adding an Auxiliary Segmentation Module (ASM), and at the same time, adding an image enhancement pipeline to assist the training in the image preprocessing stage. The powerful feature extraction capability of ConvNeXt is utilized to capture the feature of features at different scales and abstraction levels [11], and the powerful segmentation capability of Mask2Former is used to achieve the precise location and classification of land classes. The performance of my ConvNeXt-Mask2Former network model is then trained and tested for improvement on the Potsdam and Vaihingen datasets. The main contributions of this paper are as follows:

1) ConvNeXt is introduced as the backbone network, which employs a grouped convolutional structure that divides the input feature map into multiple branches for processing and performs feature fusion at the end. This design can effectively extract and make full use of the multi-scale features in the image, enhance the sensory field of the model [12], and help capture the details and contextual information in the land classification task.

2) Mask2Former is an approach that introduces the Transformer model to the semantic segmentation task. It achieves pixel-level classification and semantic segmentation by applying the Transformer module to the output of ConvNeXt. The Transformer model is able to model the relationships between pixels to capture the correlation between a pixel and multiple target regions, which solves the challenge of interdependencies in the land classification task.

3) An auxiliary segmentation module is added to the middle layer of the network, which utilizes a Fully Convolutional Neural Network (FCN) as an auxiliary head for calculating the segmentation mask for the auxiliary loss and uses the features of the middle layer of the network to assist the main network head for better learning and generalization.

4) An image enhancement pipeline is added at the stage of image preprocessing, which contains four components for data enhancement, namely scale transformation, random cropping, random flipping, and simulated lighting changes. These can be randomly applied to each training sample when the image data is loaded, which can expose the whole model to diverse samples during training, enhance the adaptability of the model to scale changes, extract rich local features, reduce the risk of overfitting, and help improve the accuracy.

II. METHOD

A. *ConvNeXt feature extraction encoder*

The model uses a simple pure convolutional network such as ConvNeXt as the backbone network, i.e. the network encoder, the main reason is that ConvNeXt doesn't use a particularly complex or innovative structure, and every detail of its network has already been used in more than one network [13], which ensures a stable architecture of the network. The network is based on the previous network step by step to modernize the architecture and finally reach the ImageNet Top-1 accuracy. It started with ResNet-50, a baseline model trained with the vision Transformer training program. Then the modernization and improvement are as follows: 1) the macro-design of the network changes the ratio of stage computation to 1:1:3:1, or 1:1:9:1 for larger models, and changes the structure of the stem; 2) the deep convolution of ResNeXt is borrowed, i.e., the idea of grouped convolution is introduced; 3) the MLP module is borrowed from the Inverted Bottleneck module from MobileNetV2 [14], which is the first one of its kind in the world. Inverted Bottleneck module, i.e., the two ends are thin and the middle is thick [15], 4) a large convolution kernel is used, and the size of the convolution kernel for deepwise convolution is changed from 3x3 to 7x7, and 5) the optimizations of micro design, including replacing ReLU [16] with GELU [17], using fewer activation functions and fewer normalization layers, replacing Bach Normalization(BN) with Layer Normalization(LN) [18], and using separate of downsampling, finally, a stable ConvNeXt is formed (Figure 1). This set of features makes ConvNeXt an effective and reliable network encoder choice.

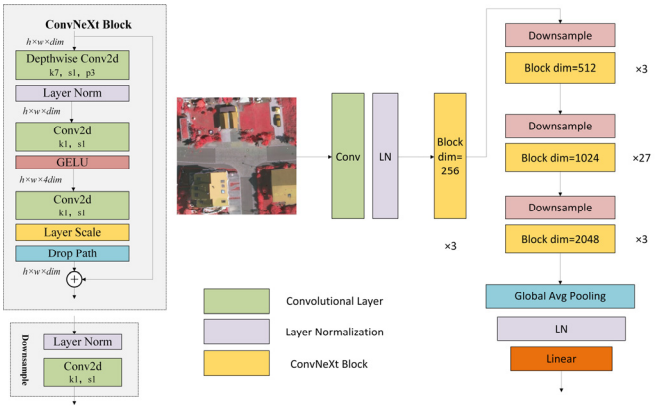


Figure 1. ConvNeXt structure diagram.

B. *Mask2Former Algorithm*

Mask2Former is built on a simple meta-framework and consists of a backbone feature extractor, a pixel decoder, and a

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Transform decoder (Figure 2). Compared to designing a specialized model for each task, Mask2Former can save up to 3 times the research effort and efficiently save computational resources. Its main improvements are as follows:

1) Masked Attention Module: Use masking attention in the Transformer decoder. The key component of the algorithm's Transformer decoder includes a mask attention operator that can extract local features by constraining the cross-attention to the foreground region of the prediction mask for each query, limiting the attention to local features centered on the predicted segments in such a way that, compared to the cross-attention used in the standard Transformer decoder, which focuses on all locations This approach gives faster convergence and better performance compared to the cross attention used in standard Transformer decoders that focus on all locations.

2) Multi-scale strategy: Use multi-scale high resolution features. High resolution images can help to improve the model results, especially for small targets. However, using all high resolution images is too expensive and increases the computational effort. Mask2Former uses a pyramid shaped decoder that uses multi-scale high resolution features, that is, successive features in the pixel decoder's feature pyramid are mapped into the Transformer decoder layer in a round robin fashion. The essence is the image pyramid. Simply put, the features of the three scales of 1/32, 1/16 and 1/8 of the original image are sent to the decoder, that is, the first layer of 1/32, the first layer of decoder is completed to get query, and the second layer is 1/16, and then the same layer is done. The Transformer decoder includes three layers, repeats L times, and finally Transformer decoder has a total of 3L layers. For each resolution, a sinusoidal position embedding and a learnable scale-level embedding are added to control the scale of each channel.

3) Optimisation improvements:

a) Switch the order of self-attention and cross-attention, because if you do self-attention between queries first, it doesn't have any information about the image and it doesn't make any sense to do so, so changing the order will make the computation more efficient.

b) Make the query features learnable as well since it is necessary to use the query feature to predict the mask at first.

c) Removed dropout completely from the decoder, as dropout affects the performance of the model and slows down training at the same time.

Sample point loss function: The mask loss is calculated at K random sampling points. Instead of calculating the loss on all points, Mask2Former samples different sets of K points based on importance sampling, which is 112×112 points assuming K is 12544. This training method saves 3 times the training memory.

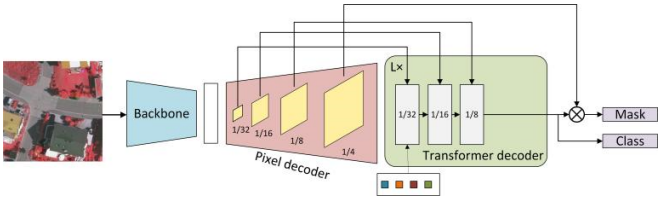


Figure 2. Mask2Former structure diagram.

C. Image enhancement pipeline

In order to enhance the model's adaptability to scale changes and reduce the risk of overfitting, I design a pipeline for adding image enhancement in the image preprocessing stage, which includes four components for data enhancement, namely scale transformation, random cropping, random flipping, and simulated illumination. Among them, scale transformation is that by changing the image scale, the model can learn the representation of the target at different sizes and improve the robustness of the model to scale changes. Random cropping is that by randomly cropping a part of the image can force the model to focus on local features of the image instead of relying only on global information, helping to reduce the model's dependence on specific regions of the image and improving the model's understanding of the object's local and contextual information. Random flipping is randomly flipping an image horizontally or vertically can increase the diversity of the data. Simulated lighting changes is that by changing the lighting conditions of an image, the model can learn to recognize objects under different light intensities and colors, allowing the model to adapt to different lighting changes. These features are essential to improve the performance of my model during training.

D. Assisted Segmentation Module (ASM)

In remote sensing image land classification task usually involves multiple land cover types which may have different representations in the feature space, in this paper, an auxiliary segmentation module is designed in the middle layer of the network, whose input is a part of the middle layer features obtained after feature extraction by the backbone network, followed by the use of a full convolutional neural network (FCN) as an auxiliary head to convert these features into a segmentation mask, which helps to improve the feature expression capability adoption, help the network to learn different types of features, promote feature fusion, and also introduce spatial information earlier, the network can then learn and maintain this spatial information earlier to further distinguish between different types of land.

In addition, the use of the full convolutional network (FCN) as an auxiliary head can also introduce additional loss functions CrossEntropyLoss and LovaszLoss, called auxiliary loss, at the same time, the final features of the backbone network are fed into the main segmentation head to generate the final segmentation mask and the main loss is calculated, and finally the total loss consists of auxiliary loss and main loss, which is used for the network's backpropagation and weights Updating. This module can also be used as a means of regularisation to prevent the model from relying too much on the last layers of the network, thus improving the generalization ability of the model and reducing overfitting. Particularly in deep networks, the

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auxiliary loss can provide an additional gradient flow to aid information transfer during optimization, helping to alleviate the gradient vanishing problem.

E. Loss Functions

Three loss functions used in the Mask2Former model are adopted in this paper, namely Classification loss, Mask loss, and DiceLoss. These three loss functions are used in combination during the model training process. The cross-entropy loss and Dice loss are complementary in terms of optimization goals. The cross-entropy loss focuses more on optimizing the classification accuracy at the pixel level, while the Dice loss focuses more on optimizing the overall shape and size of the segmentation mask. And the combination of classification loss and Dice loss helps to cope with the problem of category imbalance between land cover types, ensuring that the model is not biased towards those categories that occur more frequently. Especially when facing the task of land classification in remote sensing images, which often involves small land cover types, the Dice loss helps the model to identify these small objects because it emphasizes the overlap between the predicted and real masks. The details are as follows:

1) Classification Loss: Using the cross-entropy loss function to ensure that the model can accurately classify the category corresponding to each predicted mask, the category prediction can be optimized. It is defined as follows:

$$L_{cls} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{1}$$

Where C is the number of categories, y_i is One-Hot Encoding [19] of the real label, and $\hat{y}_i$ is the predicted probability distribution.

2) Mask Prediction Loss (MPL): The same cross entropy loss function is used to optimize the prediction of segmentation masks, which is commonly used for pixel level classification, such as the prediction of binary masks, it is defined as follows:

$$L_{mask} = 1 - \frac{2\sum(M_{pred} \cdot M_{true})}{\sum M_{pred} + \sum M_{ture}} \tag{2}$$

Where M_{pred} is the model prediction mask and M_{true} is the real mask.

3) Dice Loss: A matching loss function is used to ensure correct correspondence between the predicted mask and the true mask. Dice Loss is a commonly used metric to compare the similarity between the predicted mask and the true mask. It is defined as follows:

$$L_{dice} = 1 - \frac{2\sum_{i=1}^{N} p_i g_i + \epsilon}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} g_i + \epsilon} \tag{3}$$

Where p_i is the prediction probability of the model for pixel i to belong to the foreground category, that is, the prediction mask probability, g_i is the true binary label (0 or 1), that is, the true mask, and N is the total number of pixels, ϵ is a very small constant, used to prevent errors with a denominator of zero.

III. Experiment

A. Dataset and data pre-processing

The network model was tested on the Potsdam dataset and the Vaihingen dataset, both urban airborne remote sensing datasets produced by the International Society for Photogrammetry and Remote Sensing (ISPRS), covering a German city and its surroundings, and both labelled with six common categories: impervious surfaces, buildings, low vegetation, trees, cars, and background. Both datasets provide detailed pixel-level true labels covering a wide range of surface categories. These labels are crucial in tasks such as feature classification and building extraction. In this study, the size of each image after cropping is 512×512 pixels, and the dataset is divided into a training set, a validation set, and a test set. Potsdam, contains 3456 image pairs for the training set and 2016 image pairs for the validation set, and the test set is consistent with the validation set. Vaihingen training set contains 344 image pairs and 398 image pairs for the validation set, and the test set is consistent with the validation set. Considering the GPU memory limitation, the images in both Potsdam and Vaihingen datasets are cut to 512×512 pixel size. To avoid edge effects, I use a sliding window overlapping 50 pixels for training and testing.

B. Experimental Environment

The experimental environment for this paper is an Intel Core i7-12700H processor, Ubuntu 18.4 operating system and a GPU parallel accelerated network using NVIDIA RTX 3090 with 24G of video memory. And I used PyTorch version 1.9.0+cuda11.7 as the framework for deep learning, and the programming language was Python 3.9. AdamW was used as the optimizer, which is a variant of Adam's optimizer and introduces weight decay with a coefficient of 0.05. The initial learning rate is 0.0001 and a total of 40,000 iterations are performed, with the model information saved every 5,000 iterations and the associated evaluation metrics printed.

C. Evaluation Metrics

In order to evaluate the performance of the proposed model, different metrics are used. These indicators include Intersection over Union (IoU), Mean IoU (mIoU) , and mean Pixel Accuracy (mPA) to evaluate the performance of the model. Some of the terms used in these metrics are: p_{ii} represents the total number of pixels predicted to type i of true pixel type i, p_{ji} represents the total number of pixels predicted to type j of true pixel type i, p_{ij} represents the total number of pixels predicted to type i of true pixel type j, and k+1 represents the total number of categories, including k foreground classes and 1 background class. The IoU is a measure of the accuracy of the predicted object detection or semantic segmentation, and it calculates the ratio of the intersection of the predicted region and the real region to their union. IoU is defined as follows:

$$IoU = \frac{p_{ii}}{\sum_{j=0}^{k} p_{ij} + \sum_{j=0}^{k} p_{ji} - p_{ii}} \tag{4}$$

mIoU is the average of the IoU, which calculates and averages all classes of IoU, which provides a comprehensive evaluation of the overall performance of the model, and is defined as follows:

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$$mIoU = \frac{1}{k+1}\sum_{i=0}^{k}\frac{p_{ii}}{\sum_{j=0}^{k}p_{ij}+\sum_{j=0}^{k}p_{ji}-p_{ii}} \quad (5)$$

mPA is the average of pixel accuracy for each category, which focuses on the average of the proportion of pixels in each category that are correctly classified to all pixels in that category, and mPA is defined as follows:

$$mPA = \frac{1}{k+1}\sum_{i=0}^{k}\frac{p_{ii}}{\sum_{j=0}^{k}p_{ij}} \quad (6)$$

D. Comparison Of My Method with Other Methods

Table I shows the experimental results of different methods on the Potsdam dataset. Compared with other models, my ConvNeXt-Mask2Former model achieves the highest mIoU and mPA of 86.80% and 93.79%, respectively, and improves mIoU values by 4.55%, 2.50%, 2.17%, 2.15%, 2.15%, and 2.15%, respectively, compared with PSANet [20], FCN, DeepLabv3, DeepLabv3+ [21], DANet [22], Segformer, PSPNet [23], and UperNet [24], the mIoU values are improved by 4.55%, 2.50%, 2.17%, 2.15%, 1.83%, 1.48%, 1.11%, and 0.83%, respectively. mPA values are improved by 3.62%, 1.92%, 2%, 1.86%, 1.4%, 1.16%, 1.26%, and 0.97%, respectively. The IoU for the building category reached 94.46%, which is an improvement of 5.68% compared to PSANet.

Table I. Experimental results on Potsdam dataset.

Method	IoU (%)					mIoU (%)	mPA (%)
	Imp.surf.	Building	Low veg.	Tree	car		
PSANet	83.12	88.78	73.68	76.41	89.28	82.25	90.17
FCN	85.75	92.13	75.10	77.50	91.00	84.30	91.87
DeepLabv3	85.67	92.54	75.49	78.04	91.40	84.63	91.79
DeepLabv3+	85.97	92.14	76.21	77.96	90.95	84.65	91.93
DANet	86.26	92.61	76.24	78.80	90.95	84.97	92.39
Segformer	86.77	93.19	77.40	79.01	90.21	85.32	92.63
PSPNet	86.64	93.41	76.83	79.62	91.97	85.69	92.53
UperNet	87.71	93.81	77.32	79.65	91.34	85.97	92.82
Mine	**87.85**	**94.46**	**77.68**	**81.45**	**92.58**	**86.80**	**93.79**

a. Note: Bold indicates the best data

The experimental results of the different methods on the Vaihingen dataset in Table II show that my ConvNeXt-Mask2Former model also achieves the highest mIoU and mPA scores of 82.13% and 90.08%, respectively, compared to DANet, PSANet, FCN, Segformer, DeepLabv3, DeepLabv3+, PSPNet, and UperNet, the mIoU values are improved by 3.42%, 3.28%, 3.24%, 3.03%, 2.87%, 2.86%, 1.85%, and 1.15%, respectively. mPA values are improved by 1.93%, 2.30%, 2.38%, 1.94%, 1.99%, 2.03%, 1.46%, and 0.56%. The IoU for the building category reached 92.78%, which is an improvement of 2.63% compared to DANet. My model achieves an IoU of more than 85% for the impervious surface category on both the Potsdam and Vaihingen datasets, validating its effectiveness on both publicly available datasets.

Table II Experimental results on Potsdam dataset.

Method	IoU (%)					MIoU (%)	Mpa (%)
	Imp.surf.	Building	Low veg.	Tree	car		
DANet	84.80	90.15	70.25	79.18	70.14	78.90	88.12
PSANet	84.40	90.07	70.96	79.23	70.52	79.04	87.75
FCN	84.80	90.19	69.96	78.86	71.58	79.08	87.67
Segformer	84.79	90.53	70.78	79.81	70.55	79.29	88.11
DeepLabv3	84.67	90.77	70.51	79.48	71.80	79.45	88.06
DeepLabv3+	84.91	90.31	70.64	79.33	72.13	79.46	88.02
PSPNet	84.78	91.48	70.83	79.68	75.59	80.47	88.59
UperNet	85.87	91.57	71.54	80.52	76.36	81.17	89.49
Mine	**86.56**	**92.78**	**72.17**	**80.87**	**79.21**	**82.32**	**90.05**

a. Note: Bold indicates the best data

E. Ablation study

In order to verify the effectiveness of ASM model of the proposed method, the ablation experiment in this paper is to use the r50-Mask2Former model as the baseline, denoting the backbone network of the original model as ResNet50 [25], and subsequently replacing the Swin Transformer [26] as the backbone network, and then using the ConvNeXt as the backbone network, i.e., the feature extractor, and the Mask2Former as the decoder Training and comparison is performed for evaluating the effectiveness of the combination of ConvNeXt and Mask2Former; then ConvNeXt-Mask2Former+ASM is trained to evaluate the efficiency of the whole network model. I train on Potsdam and Vaihingen datasets respectively. Table III shows the results of ablation experiments on the Potsdam dataset.

Table III Experimental results on Potsdam dataset.

Dataset	Method	mIoU (%)	Mpa (%)
Potsdam	r50_Mask2Former	85.83	92.58
	swin_Mask2Former	85.79	92.20
	convext_Mask2Former	86.51	93.31
	convext_Mask2Former+ASM (mine)	**86.80**	**93.79**

a. Note: Bold indicates the best data

The results of the analysis show that the model with ConvNeXt as the backbone network combined with Mask2Former performs better than both the models with ResNet50 and Swin Transformer as the backbone network and the evaluation metrics mIoU and mPA are improved to 86.51% and 93.31%, respectively. After adding the ASM module again, the evaluation metrics mIoU improved by 0.29% and mPA by 0.48%, achieving the best mIoU and mPA of 86.80% and 93.79%, demonstrating the ability of the ASM module to improve model performance.

Then I trained on the Vaihingen dataset as well. Table IV shows the results of the ablation experiments on the Vaihingen dataset.

Table IV Experimental results on Vaihingen dataset.

Dataset	Method	mIoU (%)	Mpa (%)
Vaihingen	r50_Mask2Former	80.54	88.91
	swin_Mask2Former	80.36	88.71
	convext_Mask2Former	82.03	89.83
	convext_Mask2Former+ASM (mine)	**82.32**	**90.05**

a. Note: Bold indicates the best data

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The analysis results show that the model with ConvNeXt as the backbone network combined with Mask2Former also outperforms both the models with ResNet50 and Swin Transformer as the backbone network on the Vaihingen dataset, with the evaluation metrics mIoU and mPA improved to 82.03% and 89.83%, respectively. After adding the ASM module again, the evaluation metrics mIoU even improved by 0.29% and mPA by 0.22%, achieving the best mIoU and mPA, which once again proves the ability of the ASM module to improve the performance of the model.

F. Results and Discussion

The results obtained in this paper using the model combining ConvNeXt as the backbone network and Mask2Former show substantial improvements in each of the evaluation metrics, whether compared to the results obtained with ResNet50 or Swin Transformer as the backbone network combined with Mask2Former. In particular, the application of the ASM model to the ConvNeXt-Mask2Former network resulted in different degrees of improvement in the experimental results for each evaluation metric. Among them, the mIou metric improved to 86.80% and the mPA metric improved to 93.79% on the Potsdam dataset, and the mIou improved to 86.80% and the mPA improved to 93.79% on the Vaihingen dataset. It proves the effectiveness of my proposed model.

IV. Conclusion

Semantic segmentation technology has made revolutionary progress in the field of image processing, especially in remote sensing image analysis, in which the land cover classification task is to use semantic segmentation technology to distinguish different land surfaces in remote sensing images. In order to solve some deficiencies and challenges faced by semantic segmentation models at this stage in the land classification task in terms of multi-scale features of remote sensing images, the computational burden of high resolution images, and long-range dependencies. In this paper, I propose the ConvNeXt-Mask2Former network model, which is a semantic segmentation network based on the Encoder and Decoder (Encoder-Decoder) architecture, which mainly consists of ConvNeXt as an encoder, Mask2Former as a decoder, and an auxiliary segmentation module, ASM, is added in the middle layer of the network. Deep supervision in the middle layer can be provided can help the network to learn more robust feature representations, improve the training efficiency, especially help deeper network structure, and strengthen the gradient signal to reduce the gradient vanishing problem. Also, the introduction of an additional loss function can be used as a means of regularisation to prevent the model from relying too much on the last few layers of the network, thus improving the generalization ability of the model. And the image enhancement channel is added to the image preprocessing stage to assist the training. The experiments were conducted on Potsdam and Vaihingen datasets to test the efficiency of the proposed network model, and the mIoU values obtained were 86.80 and 82.32%, and the mPA values were 93.79% and 90.05%, respectively. I believe that the accuracy and robustness of semantic segmentation for land classification will be improved by the practical application of this network model.

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